Defn: The $m \times n$ "staircase" $U$ obtained by Gaussian elimination is called an <u>echelon matrix</u>. The reduced row echelon matrix $R$ has zeros above the pivots as well as below and only 1's as the pivots.

§ 3.3.

Defn: The rank of a matrix $A$ is the number of pivots and is often denoted by $r = \text{rank}(A)$.

i) $r \leq m$ and $r \leq n$.

ii) $r = m \rightarrow$ "full row rank", i.e. No zero rows in $R$.

$r = m \rightarrow$ "full column rank", i.e. No free variables.

Rank one matrix — every row is a multiple of the pivot row.

when $A \rightarrow R$.

i) $C(A) \neq C(R)$.

ii) $C(A^T) = C(R^T)$.

<u>Null space matrix</u> $N$ has $n$ rows and $n-r$ columns.

$$R = \begin{array}{c} \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{array}{l} \rightarrow r \text{ pivot rows} \\ \rightarrow m-r \text{ zero rows} \end{array} \\ \begin{array}{cc} \downarrow & \downarrow \\ r \text{ pivot} & n-r \\ \text{columns} & \text{free variables} \end{array} \end{array} \rightarrow N = \begin{bmatrix} -F \\ I \end{bmatrix} \begin{array}{l} \rightarrow r \text{ pivot variables} \\ \rightarrow n-r \text{ free variables} \end{array}$$

§ 3.4.

$Ax = b$ $\qquad [A \ b] \rightarrow [R \ d]$

Complete solution: $x = x_p + x_n$

↳ The partialar solution solves $Ax_p = b$ ↳ $Rx_p = d$

The $n-r$ special solution solve $Ax_n = 0 \rightarrow Rx_n = 0$

Full column rank ($r=n$) matrix $A$ has

i) All columns of $A$ are pivot columns

ii) no free variables or special solutions

iii) $N(A) = \{0\}$

iv) If $Ax=b$ has a solution, then it has only one solution.

Full row rank ($r=m$) matrix $A$ has

i) All rows of $A$ are pivot rows

ii) $Ax=b$ has a solution for any $b$.

iii) $C(A) = \mathbb{R}^m$

iv) There are $n-r=n-m$ special solutions.

§ 3.5

Defn: i) The columns of $A$ are <u>linearly independent</u> when the only solution to $Ax=0$ is $x=0$, i.e., $N(A)=\{0\}$.

ii) The sequence of vectors $v_1, \cdots, v_n$ are linearly independent if $c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$ only happens when $c_i = 0, i=1,2,\cdots,n$.

Defn: A set of vectors <u>span</u> a space if their linear combinations fill the space.

$C(A^T)$ — subspace of $\mathbb{R}^n$ spanned by rows of $A$.

$C(A)$ — subspace of $\mathbb{R}^m$ spanned by colums of $A$.

Defn: A <u>basis</u> for a vector space $V$ is a sequence of vectors that has two properties:

i) The vectors are linearly independent.

ii) The vectors span the space $V$.

Defn: The <u>dimension</u> of a vector space $V$ is the number of vectors in any of its basis, denoted by $\dim(V)$.

§3.6.

1. Row space of A. $C(A^T)$ is a subspace of $\mathbb{R}^n$ with dimension $r$
2. Column space of A. $C(A)$ is a subspace of $\mathbb{R}^m$ with dimension $r$
3. the Null space of A, $N(A)$ is a subspace of $\mathbb{R}^n$ with dimension $n-r$
4. The left null space of A, $N(A^T)$ is a subspace of $\mathbb{R}^m$ with dimension $m-r$.

Fundamental Theorem of Linear Algebra I

Chapter 4.

§4.1

Defn: Two vector spaces $V$ and $W$ are <u>orthogonal</u> if every vector $v$ in $V$ perpendicular to every vector $w$ in $W$, i.e.

$$v \cdot w = 0 \text{ or } v^T w = 0 \quad \text{for any } v \in V, w \in W.$$

Defn: The <u>orthognal complement</u> of $V$ contains every vector that is perpenticular to $V$, is denoted by $V^\perp$.

$$N(A) = (C(A^T))^\perp$$
$$N(A^T) = (C(A))^\perp$$

Fundamental Thorem of Linear Algebra II

$x \in \mathbb{R}^n$, $\quad x = \underbrace{x_r}_{\text{row space component}} + \underbrace{x_n}_{\text{a null space component.}}$

Any $n$ linearly independent vectors in $\mathbb{R}^n$ must span $\mathbb{R}^n$.

§4.2.

i) Projection of b onto the line through a

$$P = \left(\frac{a^T b}{a^T a}\right) a$$

Projection matrix : $\mathcal{P} = \frac{aa^T}{a^T a}$.

ii) Projection of b onto the space spanned by $a_1, a_2, \cdots, a_n$

$$A = [a_1, a_2 \cdots a_n]$$

$$P = A(A^T A)^{-1} A^T b$$

Projection matrix: $\mathcal{P} = A(A^T A)^{-1} A^T$

§4.3.

Consider $Ax = b$

Least squares solution $\hat{x}$

$$A^T A \hat{x} = A^T b$$

$\rightarrow$ make $E = \| Ax - b \|$ as small as possible.

Fitting by a straigh line.

At time $t_1, \cdots, t_m$ $(m \geq 2)$

heights $b_1, \cdots, b_m$.

best line : $y = C + Dt$.

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \rightarrow \hat{x}$$

Fitting by a parabola $y = C + Dt + Et^2$.

$$A = \begin{bmatrix} 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_m & t_m^2 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \\ E \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \longrightarrow \hat{x}$$

§4.4.

Defn: The vectors $q_1, \cdots, q_n$ are othonormal if

$$q_i^T q_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

A matrix with orthonormal columns is assigned the special letter $Q$.

$$Q^T Q = I \Rightarrow Q^T = Q^{-1}$$

$$\begin{cases} \|Qx\| = \|x\| \\ (Qx)^T (Qy) = x^T y \end{cases}$$

The Gram-Schmidt Process

$$A \rightarrow Q$$

$$A = [a_1\ a_2 \cdots a_n] \rightarrow Q = [q_1\ q_2 \cdots q_n]$$

Algorithm:

$$\left.\begin{aligned} \tilde{q}_1 &= a_1 \\ \tilde{q}_2 &= a_2 - \frac{\tilde{q}_1^T a_2}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 \\ \tilde{q}_3 &= a_3 - \frac{\tilde{q}_1^T a_3}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 - \frac{\tilde{q}_2^T a_2}{\tilde{q}_2^T \tilde{q}_2} \tilde{q}_2 \\ &\vdots \\ \tilde{q}_n &= a_n - \frac{\tilde{q}_1^T a_n}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 - \frac{\tilde{q}_2^T a_n}{\tilde{q}_2^T \tilde{q}_2} q_2 - \cdots - \frac{\tilde{q}_{n-1}^T a_n}{\tilde{q}_{n-1}^T \tilde{q}_{n-1}} \tilde{q}_{n-1} \end{aligned}\right\} \text{orthognalization}$$

$$\left. q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|}, \cdots, \quad q_n = \frac{\tilde{a}_n}{\|\tilde{q}_n\|} \right\} \text{normalization.}$$

QR factorization:

$$\underline{A = \underset{\text{orthonormal}}{Q} \underset{\text{upper triangular}}{R}} \longrightarrow \underline{R = Q^T A.}$$

# Chapter 5

## §5.1

Properties of determinant, denoted by $\det A$ or $|A|$.
$A$: $n \times n$ matrix.

(1) – (10).

In particular.

$\det(AB) = \det A \cdot \det B$ but $\det(A+B) \neq \det A + \det B$

$\det(A^{-1}) = \frac{1}{\det A}$, $\det A = \det A^T$

## §5.2.

Big formula. $\det A =$ sum of all $n!$ permutations $P = (\alpha, \beta, \cdots, \omega)$

$= \sum (\det P)\, a_{1\alpha} a_{2\beta} \cdots a_{n\omega}$ where $A = (a_{ij})_{n\times n}$.

Cofactor expansion.

$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$, $i = 1, 2, \cdots, n$.

$C_{ij} = (-1)^{i+j} \det M_{ij}$ where $M_{ij}$ is obtained by crossing out row $i$ and column $j$ from $A$.

or

$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$

## §5.3.

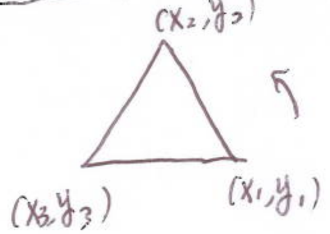
Cramer's Rule If $\det A \neq 0$, then $Ax = b$ has a solution

$$x_1 = \frac{\det B_1}{\det A}, \quad \cdots, \quad x_n = \frac{\det B_n}{\det A}$$

where $B_j$ has the $j$th column of $A$ replaced by $b$.

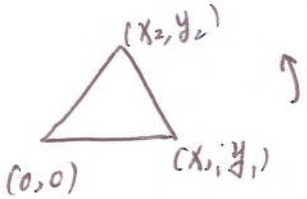
Corollary: $(A^{-1})_{ij} = \frac{C_{ji}}{\det A}$ or $A^{-1} = \frac{C^T}{\det A}$ where $C = (C_{ij})$

## Area of a triangle




right-handed complex

$$\text{Area} = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$




$$\text{Area} = \frac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$$

Volume of a parallelpiped formed by $(a_{11}, a_{12}, a_{13})$, $(a_{21}, a_{22}, a_{23})$ and $(a_{31}, a_{32}, a_{33})$ is

$$\text{volume} = |\det A| = \left|\det\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\right|$$

## Cross product

$$u \times v = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

i) $u \times v = -v \times u$

ii) $u \times u = 0$

iii) $\|u \times v\| = \|u\| \cdot \|v\| \cdot |\sin\theta|$

$\|u \cdot v\| = \|u\| \cdot \|v\| \cdot |\cos\theta|$

iv) $\text{Area}(T) = \frac{1}{2}\|u \times v\|$ where $T$ is a triangle formed by $u, v$.

## Triple product

$$(u \times v) \cdot w = \begin{vmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$